

MT PLEASANT, IA, USA

WMO: 720309

Lat: 40.947N

Lon: 91.511W

Elev: 734

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.6	2.9	-9.1	3.5	0.2	-5.9	4.1	4.8	26.8	26.3	24.5	28.0	7.2	310	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	92.9	75.4	90.4	74.6	87.6	72.6	78.8	87.6	77.2	85.8	75.4	83.5	8.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	143.8	83.8	74.5	132.6	81.6	72.8	125.0	80.0	42.9	87.7	41.0	84.5	39.4	83.9	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.0	20.5	18.5	-8.3	96.8	5.8	3.3	-12.5	99.1	-15.9	101.1	-19.2	102.9	-23.4	105.3		
			-8.1	81.8	6.4	1.7	-12.7	83.0	-16.4	83.9	-20.0	84.9	-24.7	86.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	24.8	26.7	41.4	52.1	63.3	72.6	75.0	73.1	66.6	53.4	40.6	30.0
	DBStd	20.14	12.07	11.35	11.26	9.69	8.58	5.80	5.66	5.54	8.04	9.43	9.97	10.85
	HDD50	2852	782	654	318	88	6	0	0	0	1	70	310	623
	HDD65	5878	1246	1073	737	399	138	8	2	5	75	378	732	1085
	CDD50	3495	0	1	50	150	418	679	775	715	500	176	28	3
	CDD65	1046	0	0	4	11	85	237	312	254	124	19	0	0
	CDH74	9851	0	0	26	156	780	2199	3101	2224	1191	173	1	0
	CDH80	3474	0	0	0	29	212	804	1218	770	407	34	0	0
Wind	WSAvg	8.4	9.7	9.7	10.3	10.7	9.0	7.1	5.6	5.4	6.5	8.1	9.5	9.0
Precipitation	PrecAvg	35.9	1.4	1.2	2.4	3.3	4.1	4.0	4.6	4.0	4.3	2.9	2.2	1.7
	PrecMax	59.6	3.2	3.3	6.1	10.6	11.5	14.1	13.6	11.0	12.0	7.0	6.6	5.4
	PrecMin	21.9	0.0	0.2	0.2	0.7	1.4	0.7	0.6	0.8	0.3	0.2	0.0	0.2
	PrecStd	8.1	0.9	0.8	1.5	1.8	2.4	2.3	3.0	2.6	3.0	1.9	1.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	61.5	78.6	83.6	90.2	93.4	98.4	94.0	93.5	84.4	71.9	62.0
		MCWB	46.9	52.4	63.0	62.6	70.4	75.6	77.2	74.8	73.1	68.9	58.7	57.1
	2%	DB	50.4	53.5	70.6	79.0	85.8	91.1	93.2	90.9	89.4	79.3	64.2	54.2
		MCWB	45.3	46.8	59.1	61.4	67.4	73.8	76.8	75.9	71.8	64.5	55.8	49.6
	5%	DB	44.8	45.9	63.9	73.0	81.7	88.1	90.3	87.7	85.1	73.5	60.7	49.8
		MCWB	40.5	41.0	55.7	58.3	65.9	72.2	76.1	74.1	69.5	61.7	53.1	46.0
	10%	DB	40.6	41.1	59.0	69.7	78.8	85.0	86.4	84.0	81.2	69.7	56.1	44.6
		MCWB	37.1	36.9	50.8	56.9	64.8	71.1	73.4	71.8	67.9	59.7	48.9	41.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	52.6	65.0	66.9	73.4	80.5	81.4	80.9	77.3	73.1	62.4	59.3
		MCDB	53.0	57.7	75.7	78.5	83.9	89.6	92.5	89.7	86.5	81.2	67.6	61.8
	2%	WB	45.4	47.3	61.3	63.9	71.2	77.8	79.2	78.1	74.3	69.5	58.0	50.7
		MCDB	48.8	52.4	67.0	73.7	80.5	86.1	89.0	86.0	82.8	74.3	63.2	52.4
	5%	WB	40.5	41.3	57.8	61.2	69.0	75.4	77.3	76.2	72.3	65.9	54.6	45.9
		MCDB	44.6	46.1	64.3	70.0	77.9	84.6	86.6	83.5	81.1	71.7	59.7	48.4
	10%	WB	36.8	37.1	51.9	58.4	67.1	73.3	75.6	74.4	70.4	61.2	50.3	41.2
		MCDB	40.1	40.5	57.1	67.3	75.6	81.8	84.0	81.1	78.1	67.8	55.1	44.1
Mean Daily Temperature Range	5% DB	MDBR	15.8	16.2	18.3	21.1	20.5	19.7	20.0	20.2	22.8	20.3	17.3	14.5
		MCDBR	20.5	22.3	25.6	27.8	23.9	23.3	22.7	22.9	25.0	25.7	23.5	19.5
		MCWBR	15.8	16.4	15.8	14.5	10.7	10.1	9.6	10.1	9.7	13.1	15.6	15.2
	5% WB	MCDBR	19.6	21.4	23.3	24.1	20.3	20.5	19.9	19.0	21.2	18.8	20.7	18.7
MCWBR		15.6	16.2	16.2	14.9	10.5	10.7	10.1	10.0	9.9	12.6	15.6	16.1	
Clear-Sky Solar Irradiance	taub	0.292	0.307	0.338	0.387	0.422	0.423	0.441	0.428	0.393	0.346	0.302	0.290	
	taud	2.438	2.397	2.396	2.267	2.234	2.283	2.209	2.265	2.316	2.449	2.509	2.461	
	Ebn at Noon	276	288	290	280	272	271	265	265	267	270	271	267	
	Edh at Noon	25	30	34	41	44	42	44	41	36	29	23	23	
All-Sky Solar Radiation	RadAvg	584	889	1181	1508	1766	1972	1969	1732	1481	987	664	491	
	RadStd	68	74	90	119	126	190	132	97	111	110	54	60	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page